Benchmark Diagnostics
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Insights Beyond The Diagnosis

Immunosuppression (IS) Independently Predicts Poor Outcomes in cSCC

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Tumor-related Adverse Predictors of cSCC

- Tumor size
- Perineural invasion (greater than 0.1 mm in diameter)
- Depth of invasion
- Differentiation

In immunocompetent patients, the incidence of basal cell carcinoma is greater than cSCC by a ratio of 4:1. The incidence of these lesions in the IS patients is reversed.



Patient-related Adverse Predictors of cSCC

- Immunosupression
- Location
- Age, only extreme

Most cutaneous squamous cell carcinomas (cSCCs) are successfully treated with surgical modalities.

However, certain immunosuppressive (IS) states may lead to poor outcomes independent of known histological and site-specific predictors of aggressive behavior.

Bone marrow and solid organ transplants, as well as hematologic malignancies, resulted in adverse outcomes

Causes of immunosuppression that did not lead to poor outcomes:

Patients with HIV or drug- induced immunosuppression.

Poor Patient Outcomes in this study included:

- Recurrence
- Nodal metastasis
- Distent metastasis
- cSCC-specific mortality

Pearls

- Treatment of cSCC in IS patients is often multifaceted and requires closer surveillance.
- Adjusting immunosuppressive regimens to minimize their carcinogenic potential is generally advised
- A retrospective study demonstrated that 70% of cutaneous squamous cell carcinoma in situ (cSCC-IS) cases in immunocompetent patients cleared after biopsy, but only 47% of cases were cleared in IS patients.
- 5-fluorouracil is reported to be the most effective at reducing the number of actinic keratosis lesions in IS cases
- IS patients tend to have more aggressive disease characteristics; Mohs micrographic surgery appears to be effective, as it is generally the AAD recommendation for higher-risk SCC.
- Radiation therapy and chemotherapy may be used only if surgery is not an option. Their adjuvant role has not been well studied.
- Immunotherapy's safety and efficacy are also being evaluated

Li S, Townes T, Na'ara S. Current Advances and Challenges in the Management of Cutaneous Squamous Cell Carcinoma in Immunosuppressed Patients. Cancers.

Immunosuppression as an Independent Risk Factor for Poor Outcomes in Cutaneous Squamous Cell Carcinoma: A Prospective Study. Rosenthal A, Conde G, et al: Dermatol Surg; 2025;51 (September): 852-858.

"Questions or cases to discuss? We're always happy to review."

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